

ERP Pipeline Flow Documentation

Workflow: Sales Order (SO) → Bill of Materials (BOM) → Material Requirements Planning (MRP) → Purchase Requisition (PR) → RFQ / Convert PO → Purchase Order (PO) → Goods Receipt Note (GRN) → Stock Update → Job Order (JO) → Material Issue

Document Purpose: This document serves as a complete reference guide detailing how business processes, database structures, and backend CodeIgniter 3 codebase controllers tie together in the manufacturing and procurement pipeline.

1. The Pipeline Steps

Step 1: Sales Order (SO) & Bill of Materials (BOM) Setup

Goal: Establish sales demand and define raw components needed to build the finished goods.

Database Tables:

- `sameepaccounting_salesorder_total` — Sales Order Headers
- `sameepaccounting_salesorder` — Sales Order Finished Goods Items
- `sameepaccounting_bom_total` — BOM Headers
- `sameepaccounting_bom` — BOM Component breakdowns

Key Controllers/Models:

- `application/controllers/BomController.php`

Step 2: Running MRP & Explosion

Goal: Calculate the raw component demands for the Sales Order and evaluate inventory shortages.

Calculations:

- **Gross Required Quantity:** $\text{BOM Component Quantity} * \text{Sales Order Item Quantity}$
- **Pending Required Quantity:** $\text{Max}(0, \text{Gross Required} - \text{Already Issued Quantity})$
- **Shortage:** $\text{Max}(0, \text{Pending Required} - \text{Available Stock})$

Key Controllers/Models:

- `application/controllers/MaterialIssueController.php`
→ `run_sales_order_mrp()`
- `application/models/Material_issue_model.php` →
`get_sales_order_mrp_data()`

Step 3: Generate Purchase Requisition (PR) & Approval

Goal: Generate a requisition for all items with shortages and obtain managerial sign-off.

Database Tables:

- `sameepaccounting_purchase_requisition` — Header record with `location_id_fk` and status (Pending)
- `sameepaccounting_purchase_requisition_items` — Items, quantities, and MRP context specs

Key Controllers/Models:

- `application/controllers/MaterialIssueController.php`
→ `ajax_generate_pr_from_mrp()`
- `application/controllers/RequisitionController.php` →
`approval_dashboard()` (PR Approval Dashboard)

Step 4: Procurement Pathway Options

Once a PR is approved by the Site Incharge, two procurement routes become available:

Pathway A: Direct PO Conversion

- Click **Convert to PO** on the approved PR.
- The system redirects to the PO create screen and auto-populates the grid with the PR items via an AJAX call.
- **Files:** `application/controllers/SupplierController.php`
→ `create_purchase_order()` , `get_pr_items_for_po()`

Pathway B: RFQ & Comparison

- Click **Convert to RFQ** to issue a request for quote to multiple vendors.
- Add received quotations from suppliers.
- Use the comparison screen to review **L1, L2, L3** pricing and convert the chosen vendor quotation directly to a PO.
- **Files:** `application/controllers/RFQController.php` ,
`application/controllers/SupplierController.php` →
`convert_rfq_to_po()`

Step 5: Goods Receipt Note (GRN) & Stock Update

Goal: Record the receipt and inspection of purchased goods and increase inventory stock.

Stock Update Logic:

When the GRN is created and passes quality inspection, checking the **"Update Stock"** checkbox executes the stock addition:

```
UPDATE sameepaccounting_inventory SET stock = stock +
accepted_quantity WHERE item_name = product_name;
```

Key Controllers/Models:

- `application/controllers/GrnController.php` →
`create_grn()` , `add_grn()`
- `application/models/Grn.php` →
`update_stock_after_inspection()`

Step 6: Job Order (JO) & Material Issue (Stock Deduction)

Goal: Start the production job order and release raw materials from stock to manufacturing.

Stock Deduction Logic:

When creating a Material Issue slip referencing the Job Order, the system checks stock availability and immediately deducts stock:

```
UPDATE sameepaccounting_inventory SET stock = stock -  
issued_quantity WHERE inventory_id = inventory_id_fk;
```

A stock ledger record is generated for auditing and balance tracing.

Key Controllers/Models:

- `application/controllers/JobOrderController.php` (JO Master setup)
- `application/controllers/MaterialIssueController.php`
→ `create()` (Material Issue Slip creation)
- `application/models/Material_issue_model.php` →
`create_issue_slip()`, `update_inventory_stock()`